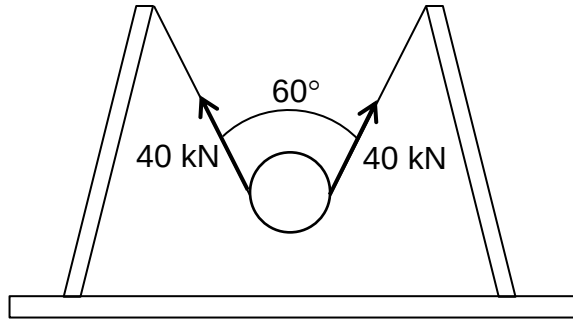


- 6** In reverse bungee jumping, passengers in a spherical cage are pulled upwards by a pair of elastic cords attached to the side of the cage as shown.

The weight of the passengers and the cage is W .



What is the maximum value of W if the passengers and the cage are flung upwards with an acceleration of g ?

- A** 20 kN **B** 35 kN **C** 46 kN **D** 69 kN

B Resultant upward force = $2(40\text{kN} \cos 30^\circ) - W = W$
 $80\text{kN} \cos 30^\circ = 2W$
 $W = 40\text{kN} \cos 30^\circ = 35 \text{ kN}$

7 A 2.0 kg block slides down a plane that is inclined at 25° with the horizontal, at a